FINAL EXPLANATION: MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: MATHEMATICS GRADE 10 – SUB-STRAND 3.2: ROTATION	
Student Assessment Document	
Student Name	____
Class	____
Date	____

INSTRUCTIONS FOR STUDENTS
Your Final Explanation is your chance to show everything you have learned in this unit by answering the Driving Question fully and clearly. Read the Driving Question carefully, then write a detailed explanation using the four sections below as your guide. Your answer should connect back to the Ngong Hills wind turbine phenomenon you investigated at the start of the unit. Use correct mathematical vocabulary, include worked examples with coordinates, and support every claim with evidence or reasoning from your lessons. Work through each section in order — each one builds on the last. You will be assessed using the rubric at the bottom of this document.

Section 1: What Is Rotation — and What Does the Ngong Hills Turbine Show Us?	
Explain what a rotation is as a geometric transformation. In your answer: define rotation using correct mathematical vocabulary (centre of rotation, angle of rotation, direction of rotation); explain what stays the same and what changes when a shape is rotated; and connect your explanation directly to the Ngong Hills wind turbine — identify the centre of rotation on the turbine, describe what is preserved as the blades spin, and explain why the overall shape of the three-blade assembly looks identical no matter where the blades stop.	A rotation is a geometric transformation that turns every point of a shape through a fixed angle about a fixed point called the centre of rotation. The centre of rotation is the only point that does not move. During a rotation, the size, shape, and side lengths of the object are all preserved — only the position and orientation of the shape change. This means rotation is an isometric transformation: the object and its image are always congruent. On the Ngong Hills wind turbine, the centre of rotation is the central hub where all three blades meet. As the turbine spins, every point on every blade traces a circular arc around this hub. The length of each blade, the angles between the blades, and the overall size of the rotor are all preserved throughout the rotation. This is why the turbine always looks the same — at any stopping position, the three-blade assembly maps perfectly onto itself because each blade simply moves into the position previously occupied by another blade. The shape has rotational symmetry, which we will explore further in Section 2.

Section 2: Rotational Symmetry and Why Three Blades Work So Well	
Explain the concept of rotational symmetry and apply it to the turbine. In your answer: define the order of rotational symmetry and the angle of rotational symmetry; calculate the order and minimum angle of rotational symmetry for the three-blade turbine; explain why a person watching the turbine cannot tell whether it has rotated by the minimum angle — even though the blades have clearly moved through space; and state what would happen to this symmetry if the turbine had two blades or four blades instead.	A shape has rotational symmetry if it maps onto itself after a rotation of less than 360°. The order of rotational symmetry is the number of times the shape looks identical to its starting position during one full 360° turn. The minimum angle of rotational symmetry (also called the angle of rotational symmetry) is found using the formula: Angle of rotational symmetry = 360° ÷ order The Ngong Hills turbine has three blades equally spaced around the hub. This means it maps onto itself every time it rotates by 360° ÷ 3 = 120°. The order of rotational symmetry is therefore 3, and the minimum angle is 120°. When the turbine rotates exactly 120°, Blade A moves into the position of Blade B, Blade B moves into the position of Blade C, and Blade C moves into the position of Blade A. Because all three blades are identical in length and shape, an observer cannot distinguish the new position from the original — this is exactly why the turbine always looks the same. If the turbine had two blades, the order of rotational symmetry would be 2 and the minimum angle would be 180°. If it had four blades, the order would be 4 and the minimum angle would be 90°. Three blades are commonly chosen in real wind farms like Ngong Hills because this balance of symmetry gives the most efficient and stable rotation in typical wind conditions.

Section 3: The Mathematics of Rotation — Mapping Points Using Rules	
Show how you can use coordinate geometry to predict exactly where any point on a turbine blade will land after a rotation. In your answer: state the coordinate rotation rules for rotations of 90°, 180°, and 270° anticlockwise about the origin; apply the 120° rotation concept by working through a clear numerical example with at least two points (you may use a tip of a turbine blade as your context); explain the meaning of the image and the pre-image; and show that the distance from the centre of rotation to each point is preserved before and after the rotation.	When a point is rotated about the origin (0, 0), we can use the following rules: – 90° anticlockwise: (x, y) → (−y, x) – 180° (either direction): (x, y) → (−x, −y) – 270° anticlockwise (= 90° clockwise): (x, y) → (y, −x) The original point is called the pre-image, and the point it maps to after rotation is called the image. Let us model a simplified turbine blade. Suppose one blade tip is at point A(6, 0) and a midpoint along the same blade is at point B(3, 0), with the hub at the origin. After a rotation of 180° anticlockwise about the origin: – A(6, 0) → A′(−6, 0) using the rule (x, y) → (−x, −y) – B(3, 0) → B′(−3, 0) We can verify that distances from the centre are preserved: – Distance OA = $\sqrt{6^2 + 0^2}$ = 6 units; Distance OA′ = $\sqrt{(-6)^2 + 0^2}$ = 6 units ✓ – Distance OB = $\sqrt{3^2 + 0^2}$ = 3 units; Distance OB′ = $\sqrt{(-3)^2 + 0^2}$ = 3 units ✓ This confirms that rotation preserves distances from the centre of rotation — every point on a blade stays exactly as far from the

	hub as it was before the rotation. This is why the blade does not bend, stretch, or change shape as it sweeps through the air above Ngong Hills.

Section 4: Finding the Centre and Angle of Rotation from an Object and Its Image	
Explain how you can work backwards — given an object and its rotated image, determine the centre of rotation and the angle of rotation. In your answer: describe the method of using perpendicular bisectors of line segments joining corresponding points to locate the centre of rotation; describe how to measure or calculate the angle of rotation using the centre you have found; work through a brief example to illustrate the method; and explain how an engineer at the Ngong Hills wind farm could use this technique in a real context (for example, checking that a repaired blade is correctly repositioned).	To find the centre of rotation when you are given an object and its image, follow these steps: Step 1 — Join corresponding points: Draw line segments connecting each vertex (or key point) of the object to its matching vertex on the image. For example, connect pre-image point P to its image P′, and pre-image point Q to its image Q′. Step 2 — Construct perpendicular bisectors: Draw the perpendicular bisector of segment PP′ and the perpendicular bisector of segment QQ′. A perpendicular bisector passes through the midpoint of a segment at a 90° angle. Step 3 — Identify the centre: The two perpendicular bisectors will intersect at exactly one point — this intersection is the centre of rotation. Step 4 — Measure the angle: Draw line segments from the centre of rotation to any pre-image point and to its corresponding image point. The angle between these two segments, measured at the centre, is the angle of rotation. The direction (clockwise or anticlockwise) is determined by observing which way the image has turned relative to the object. Example: Suppose a blade tip moves from P(4, 0) to P′(0, 4). The midpoint of PP′ is (2, 2) and the perpendicular bisector of PP′ passes through the origin (0, 0). Checking with a second point confirms the centre is the origin. The angle ∠POP′ = 90° anticlockwise. Real-world application: If an engineer at Ngong Hills replaces a damaged blade and needs to confirm it is fixed at exactly 120° from the neighbouring blades, they can treat the hub as the centre of rotation, measure the angular positions of key points on each blade, and verify using this method that the rotation angle between adjacent blades is exactly 120°. Any deviation would indicate the blade is misaligned.

Section 5: Full Answer to the Driving Question	
Now bring everything together. Write a complete, well-organised answer to the Driving Question: 'Why do the blades of the Ngong Hills wind turbine always look the same no matter how far they rotate — and how can we use the mathematics of rotation to predict exactly where any point on the blade	The Ngong Hills wind turbine blades always look the same no matter how far they rotate because the three-blade assembly has rotational symmetry of order 3. This means that after every rotation of exactly 120° (= 360° ÷ 3) about the central hub, each blade moves precisely into the position previously occupied by the next blade. Because all three blades are identical in size and shape, an observer cannot detect any difference — the image is indistinguishable from the object. This is not a coincidence; it is a direct consequence of the mathematical properties of rotation. Rotation preserves length, angle size, and shape (it is an isometric transformation), so the overall rotor shape — its size, its symmetry, its appearance — is completely unchanged after any multiple of 120°. We can use the mathematics of rotation to predict exactly where any point on a blade will land. For rotations about the origin, we apply coordinate transformation rules: 90° anticlockwise maps (x, y) to (−y, x); 180° maps (x, y) to (−x, −y); and 270° anticlockwise

will land?' Your answer must reference all four previous sections, use correct mathematical vocabulary throughout, and clearly explain both parts of the Driving Question — the 'why' (rotational symmetry and preserved properties) and the 'how' (coordinate rules and geometric methods). Aim for at least one paragraph per part of the question.	maps (x, y) to (y, −x). For any angle, the key property is that the distance from every point to the centre of rotation is preserved — meaning every blade tip, midpoint, and fixing bolt on the turbine traces a perfect circle around the hub. If you know a point's starting coordinates and the angle of rotation, you can calculate its exact landing position. Finally, if you are given only an object and its image — for example, a blade before and after repositioning — you can work backwards to find the centre and angle of rotation by constructing perpendicular bisectors of the segments joining corresponding points. Their intersection gives the centre, and the angle between the pre-image and image rays from that centre gives the rotation angle. Together, these tools — symmetry, coordinate rules, and the perpendicular bisector method — give us a complete mathematical explanation of why the Ngong Hills turbine looks the same at every position, and the power to predict and verify the exact position of any point on its blades.

RUBRIC			
Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**
Understanding of Rotation as a Transformation	Accurately and completely defines rotation using all key vocabulary (centre, angle, direction, isometry). Clearly identifies what is preserved and what changes, with explicit and correct reference to the Ngong Hills turbine context.	Defines rotation with most key vocabulary and correctly identifies preserved properties. Reference to the turbine context is present but may lack full precision or completeness.	Shows a partial or incomplete understanding of rotation. Misses one or more key properties or vocabulary terms. Connection to the turbine phenomenon is vague or absent.
Rotational Symmetry — Order and Angle Calculation	Correctly calculates the order of rotational symmetry (3) and the minimum angle (120°) using the formula, with clear working shown. Accurately explains why the three-blade arrangement produces identical-looking positions and correctly generalises to other blade numbers.	Correctly identifies the order and angle with working shown. Explanation of why the turbine looks the same is mostly complete. Generalisation to other blade numbers may be partially correct.	Attempts to calculate order or angle but contains errors in formula use or reasoning. Little or no explanation of how symmetry produces identical-looking positions.
Application of Coordinate Rotation Rules	States all three standard rotation rules (90°, 180°, 270°) correctly, applies them accurately to at least two points in a relevant example, and verifies that distances from the centre of rotation are preserved with clear calculation.	States most rotation rules correctly and applies them to at least one point with mostly accurate working. Distance preservation is mentioned but may not be fully verified with calculation.	States rotation rules with errors or applies them incorrectly. Little or no attempt to verify distance preservation. Example may be absent or incorrect.
Finding Centre and Angle of Rotation from Object and Image	Clearly and correctly describes the perpendicular bisector method step by step. Accurately explains how to determine the angle of rotation. Provides a correct illustrative example and gives a convincing real-world application linked	Describes the perpendicular bisector method with mostly correct steps. Angle determination is explained with minor gaps. Real-world application is present but may be general rather than turbine-	Shows a partial or confused description of the method for finding the centre of rotation. Steps are incomplete or contain significant errors. Real-world application is absent or unclear.

	to the Ngong Hills turbine context.	specific.	
Quality of Final Driving Question Answer	The final answer is well-organised, addresses both parts of the Driving Question fully (the 'why' and the 'how'), integrates all four preceding sections, uses correct mathematical vocabulary consistently, and demonstrates a sophisticated and coherent understanding of rotation as a whole.	The final answer addresses both parts of the Driving Question and references most preceding sections. Mathematical vocabulary is mostly correct. The overall argument is clear but may lack full integration or depth in one area.	The final answer addresses only one part of the Driving Question or is poorly organised. References to preceding sections are weak or missing. Mathematical vocabulary is limited or used incorrectly.